

Challenge (Habanero)

- Q1.** Does the function $f(x) = x^2 + 1$ have an inverse?
- (A) Yes
(B) No
(C) Only for $x > 0$
(D) Only for $x < 0$
(E) Depends on the domain
- Q2.** Find the inverse of the function $f(x) = 2x - 3$. What is $f^{-1}(x)$?
- (A) $\frac{x+3}{2}$
(B) $\frac{x-3}{2}$
(C) $2x + 3$
(D) $\frac{x}{2} - 3$
(E) $\frac{x}{2} + 3$
- Q3.** The fuel consumption of a hiking group is given by $f(x) = 0.5x + 2$, where x is the distance traveled in miles. Find the inverse function $f^{-1}(x)$. What does it represent?
- (A) Distance traveled given fuel consumption, $f^{-1}(x) = 2x - 4$
(B) Fuel consumption given distance traveled, $f^{-1}(x) = 2x - 4$
(C) Distance traveled given fuel consumption, $f^{-1}(x) = 2x + 4$
(D) Time taken given fuel consumption, $f^{-1}(x) = x - 4$
(E) Fuel consumption given distance traveled, $f^{-1}(x) = x + 4$
- Q4.** If $f(x) = x^2$ and $g(x) = \sqrt{x}$, which statement is true?
- (A) $f(g(x)) = x$ and $g(f(x)) = x$
(B) $f(g(x)) = x$ but $g(f(x)) \neq x$
(C) $g(f(x)) = x$ but $f(g(x)) \neq x$
(D) $f(g(x)) \neq x$ and $g(f(x)) \neq x$
(E) $f(g(x)) = x^2$ and $g(f(x)) = \sqrt{x}$
- Q5.** The function $f(x) = 3x - 2$ is one-to-one. What is the domain of the inverse function $f^{-1}(x)$?
- (A) All real numbers
(B) All integers
- (C) All rational numbers
(D) $x \geq -2$
(E) $x \leq 2$
- Q6.** The population of a certain animal species after x years is given by $f(x) = 200(1.05)^x$. Find the inverse function and interpret its meaning.
- (A) $f^{-1}(x) = \log_{1.05}(\frac{x}{200})$, represents the number of years for the population to reach x animals
(B) $f^{-1}(x) = \log_{1.05}(\frac{x}{200})$, represents the number of animals after x years
(C) $f^{-1}(x) = \frac{1}{1.05}(\frac{x}{200})$, represents the number of years for the population to reach x animals
(D) $f^{-1}(x) = \log_{1.05}(\frac{200}{x})$, represents the number of years for the population to reach x animals
(E) $f^{-1}(x) = \frac{1}{1.05}(\frac{200}{x})$, represents the number of animals after x years
- Q7.** The altitude of a hiker is given by $f(x) = -2x + 1000$, where x is the horizontal distance traveled in kilometers. Find the inverse of $f(x)$. What does the inverse represent?
- (A) $f^{-1}(x) = -\frac{x-1000}{2}$, represents the horizontal distance for a given altitude
(B) $f^{-1}(x) = -\frac{x-1000}{2}$, represents the altitude for a given horizontal distance
(C) $f^{-1}(x) = -\frac{x+1000}{2}$, represents the horizontal distance for a given altitude
(D) $f^{-1}(x) = \frac{x-1000}{2}$, represents the horizontal distance for a given altitude
(E) $f^{-1}(x) = \frac{x+1000}{2}$, represents the altitude for a given horizontal distance
- Q8.** If $f(x) = x^3 + 1$ and $g(x) = \sqrt[3]{x-1}$, which of the following is true?
- (A) $f(g(x)) = x$ and $g(f(x)) = x$
(B) $f(g(x)) = x$ but $g(f(x)) \neq x$
(C) $g(f(x)) = x$ but $f(g(x)) \neq x$
(D) $f(g(x)) \neq x$ and $g(f(x)) \neq x$
(E) $f(g(x)) = x^3$ and $g(f(x)) = \sqrt[3]{x}$

Open Ended Questions (FRQ)

- Q9.** Consider the function $f(x) = x^2 - 4x + 3$. Find the inverse function $f^{-1}(x)$. Then use the inverse to find the value of x such that $f(x) = 5$. Show all work and justify your answer.
- Q10.** A water tank can be filled at a rate of $f(x) = 2x + 1$ gallons per minute, where x is the number of minutes. Find the inverse function $f^{-1}(x)$ and interpret its meaning. Then, if the tank can hold 1000 gallons, how long will it take to fill the tank? Show all work and justify your answer.

Chef's Tips

- Read each question carefully before answering.
- Use the given formulas and equations to solve the problems.
- Show all work and justify your answers for free-response questions.